

Simulating Fluid Dynamics of a Formula SAE Car

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Motivation: I have always been wanting to perform a fluid flow analysis of a Formula racecar. This project provided me with an opportunity to showcase my skills and deepen my understanding of the fluid flow behavior around a moving object. My experience showed me that computer modelling is paramount in providing accurate data and simulation models for the fluid flow dynamics around a body with known boundary conditions.

Project Description: The Society of Automotive Engineers (SAE) created the challenge of performing the simulation of a Formula SAE car. The assignment consisted of developing an accurate fluid flow simulation while taking into account the mesh sensitivity analysis with the quantification of y^+ and the effect of different turbulence models on the flux reports.

Testing: We started by modelling the FSAE car and reducing the complex geometries to minimize the number of computational cells and save on computational resources. The next step was to create an enclosure to model the fluid flow region.

Table 1

Mesh sensitivity	y^+
848589 cells	40.555655
1.5 million cells	62.348838
2.1 million cells	62.684907

The meshing was done on the model by applying the different cell zone sizes to the near-field and far-field regions of the computational volume. Table 1 and 2 shows the mesh sensitivity analysis and the effect of different turbulent models with the quantification of y^+ . From the computational results shown in the table, it's clear that there is an increase in the y^+ value with the increasing number of cells for this model from 848589 cells to approx. 1.5 million, but there is not much significant difference between the y^+ value obtained for the 1.5 million and 2.1 million mesh models. A high value of y^+ (typically greater than 30) indicates that the flow is in the log layer and a high-Reynolds-number model can be used to accurately predict the flow behavior indicating higher accuracy of the mesh. Therefore, the value of y^+ is used to choose an appropriate turbulence model and to determine the accuracy of the near-wall flow prediction in a fluid flow simulation. The model with 848589 computational cells corresponds to the same y^+ value of the k-omega turbulence model as shown in table 2 given below indicating that k-omega is the most accurate turbulence model to be used in this case.

Table 2

Turbulence Models	y^+
k-omega	40.555655
k-epsilon	41.668862
transition k-kl omega	52.01051

Challenges: The most challenging part of this project was to try and manipulate the mesh settings to generate a uniform mesh with good orthogonal quality. Performing multiple mesh sensitivity analyses with different element sizes was time-consuming. Furthermore, performing the analysis of the FSAE car with different turbulent models was an added burden.

Meaning: Now that I have performed the fluid flow simulation, I can see many applications of it in the industry and academia using computer modelling. For example, I can apply this idea to simulate the fluid flow analysis of a much bigger and more complicated body such as an aircraft. But this obviously would be more complicated with a much deeper understanding of the boundary conditions and better skills with the meshing of certain complex geometries and fine-tuning the necessary parameters.

Possible Improvements: I would suggest that it would be easier and less burdensome for someone who is an intermediate in fluid flow simulations to get familiarized with ANSYS Fluent by performing simulation on a much simpler object to understand and determine the effects of different meshing parameters and fluid flow models. Although this project is challenging and time-consuming it is very interesting.

